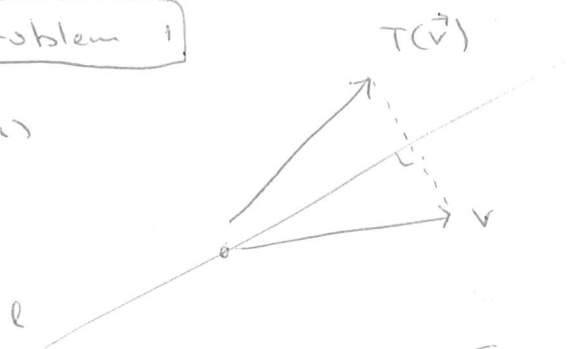


1MA-188

Exam 2025-04-14

Problem 1

(a)



From picture, it is clear that $\vec{v} + T(\vec{v})$ must be parallel to l

$$\text{So let } \bar{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + T\left[\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}\right]$$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix}$$

Equation:
$$\begin{cases} x = -2t \\ y = 0 \\ z = 4t \end{cases}$$

(b) Standard matrix of $T(\bar{x}) = 2\text{proj}_{\bar{w}}(\bar{x}) - \bar{x}$

is
$$\frac{2\bar{w}\bar{w}^T}{\|\bar{w}\|^2} - I = \frac{2}{20} \begin{pmatrix} 4 & 0 & -8 \\ 0 & 0 & 0 \\ -8 & 0 & 16 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -3/5 & 0 & -4/5 \\ 0 & -1 & 0 \\ -4/5 & 0 & 3/5 \end{pmatrix}$$

Problem 2

$$(a) \left(\begin{array}{ccc|ccc} x & \sqrt{1-x^2} & 0 & 1 & 0 & 0 \\ \sqrt{1-x^2} & -x & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \right) \cdot \frac{1}{x} \quad (\text{since } x > 0, \text{ this is ok})$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & \frac{\sqrt{1-x^2}}{x} & 0 & \frac{1}{x} & 0 & 0 \\ \sqrt{1-x^2} & -x & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \right) - \sqrt{1-x^2} \cdot I$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & \frac{\sqrt{1-x^2}}{x} & 0 & \frac{1}{x} & 0 & 0 \\ 0 & -x - \frac{1-x^2}{x} & 0 & -\frac{\sqrt{1-x^2}}{x} & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & +1 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & \frac{\sqrt{1-x^2}}{x} & 0 & \frac{1}{x} & 0 & 0 \\ 0 & -\frac{1}{x} & 0 & -\frac{\sqrt{1-x^2}}{x} & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & +1 \end{array} \right) \begin{array}{l} + II \cdot \sqrt{1-x^2} \\ \cdot (-x) \\ \cdot (-1) \end{array}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{x} - \frac{1-x^2}{x} & \sqrt{1-x^2} & 0 \\ 0 & 1 & 0 & \sqrt{1-x^2} & -x & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & x & \sqrt{1-x^2} & 0 \\ 0 & 1 & 0 & \sqrt{1-x^2} & -x & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\text{so } A^{-1} = \left(\begin{array}{ccc} x & \sqrt{1-x^2} & 0 \\ \sqrt{1-x^2} & -x & 0 \\ 0 & 0 & -1 \end{array} \right)$$

$$(b) A^{-1} = A$$

$$\text{So } A^2 = I \Rightarrow A^4 = I \Rightarrow \dots \Rightarrow A^{2024} = I \Rightarrow A^{2025} = A$$

Problem 3

(a) $\frac{(x-y)^3}{x^2+y^2}$ is a rational function that is continuous everywhere except where denominator is zero, i.e. $(0,0)$

Need to check $(0,0)$ separately:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{(x-y)^3}{x^2+y^2} = \lim_{\substack{r \rightarrow 0 \\ \text{uniformly in } \theta}} \frac{(r \cos \theta - r \sin \theta)^3}{r^2}$$

= 0 since

$$0 \leq \left| \frac{r^3 (\cos \theta - \sin \theta)^3}{r^2} \right| \leq r \cdot (1+1)^3 = 8r$$

$\downarrow$ $\downarrow$ $\downarrow$
 0 0 0

by Squeeze theorem

Since $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 = f(0,0)$, f is

continuous at $(0,0)$ as well

So f is continuous everywhere

(b)

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{(h-0)^3}{h^2+0^2} - 0}{h}$$

by definition $\nearrow$

$$= \lim_{h \rightarrow 0} \frac{h^3}{h^3} = 1$$

(c) If $(x,y) \neq (0,0)$, then we just differentiate directly:

$$f_x(x,y) = \frac{\partial}{\partial x} \left(\frac{(x-y)^3}{x^2+y^2} \right) = \frac{3(x-y)^2(x^2+y^2) - 2x(x-y)^3}{(x^2+y^2)^2}$$

(d) Need to compute $\lim_{(x,y) \rightarrow (0,0)} f_x(x,y)$ and see

if it coincides with $f_x(0,0) = 1$

$$\frac{3(r \cos \theta - r \sin \theta)^2 \cdot r^2 - 2r \cos \theta (r \cos \theta - r \sin \theta)^3}{r^4}$$

$$\lim_{(x,y) \rightarrow (0,0)} f_x(x,y) = \lim_{\substack{r \rightarrow 0 \\ \text{uniformly in } \theta}}$$

$$= \lim_{\substack{r \rightarrow 0 \\ \text{uniformly in } \theta}} \left(3(\cos \theta - \sin \theta)^2 - 2 \cos \theta (\cos \theta - \sin \theta)^3 \right) - \text{depends on } \theta$$

So limit doesn't exist

(e) $D_{\vec{u}} f(0,0) = \lim_{h \rightarrow 0} \frac{f(u_1 h, u_2 h) - f(0,0)}{h}$

by definition

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{(u_1 h - u_2 h)^3}{u_1^2 h^2 + u_2^2 h^2} \right) - 0}{h} = \frac{(u_1 - u_2)^3}{u_1^2 + u_2^2} = (u_1 - u_2)^3$$

↑
since $(u_1^2 + u_2^2) = |\vec{u}|^2 = 1$

Problem 4

$F(x, y, z)$

(a)
$$\begin{cases} x^2 + y^3 + z^4 = 3 \\ xy + xz + yz = 3 \end{cases}$$

$G(x, y, z)$

IFT:
$$\frac{\partial(F, G)}{\partial(x, y)} = \det \begin{pmatrix} F_x & F_y \\ G_x & G_y \end{pmatrix} = \det \begin{pmatrix} 2x & 3y^2 \\ y+z & x+z \end{pmatrix}$$

At $(1, 1, 1)$:

$$\frac{\partial(F, G)}{\partial(x, y)} = \det \begin{pmatrix} 2 & 3 \\ 2 & 2 \end{pmatrix} = 4 - 6 = -2 \neq 0$$

So $x = x(z)$
 $y = y(z)$ works

(b)
$$\frac{\partial x}{\partial z} = - \frac{\frac{\partial(F, G)}{\partial(z, y)}}{\frac{\partial(F, G)}{\partial(x, y)}} = - \frac{\det \begin{pmatrix} F_z & F_y \\ G_z & G_y \end{pmatrix}}{-2} = - \frac{\det \begin{pmatrix} 4z^3 & 3y^2 \\ x+y & x+z \end{pmatrix}}{-2}$$

$$= - \frac{\det \begin{pmatrix} 4 & 3 \\ 2 & 2 \end{pmatrix}}{-2} = 1$$

$$\frac{\partial y}{\partial z} = - \frac{\frac{\partial(F, G)}{\partial(x, z)}}{\frac{\partial(F, G)}{\partial(x, y)}} = - \frac{\det \begin{pmatrix} F_x & F_z \\ G_x & G_z \end{pmatrix}}{-2} = - \frac{\det \begin{pmatrix} 2x & 4z^3 \\ y+z & x+y \end{pmatrix}}{-2}$$

$$= - \frac{\det \begin{pmatrix} 2 & 4 \\ 2 & 2 \end{pmatrix}}{-2} = -2$$

Problem 5

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$= f_x \cdot 2s + f_y \cdot 2t$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$= f_x (-2t) + f_y \cdot 2s$$



So new PDE is

$$s \cdot (f_x \cdot 2s + f_y \cdot 2t) - t \cdot (-2t f_x + 2s f_y) = 2s^4 - 2t^4$$

$$(s^2 + t^2) f_x = s^4 - t^4 = (s^2 - t^2)(s^2 + t^2)$$

$$f_x = s^2 - t^2 = x$$

$$f_x(x, y) = x$$

Integrate with respect to x :

$$f(x, y) = \frac{x^2}{2} + g(y) \text{ for any function } g \text{ of } y$$

Answer: $f(s, t) = \frac{(s^2 - t^2)^2}{2} + g(2st)$

Problem 6

$$T(x, y) = x^4 + 2y^2 + 1 \rightarrow \min$$

$$\text{subject to } \underbrace{y - x^2 + 3 = 0}_{g(x, y)}$$

$$\text{Lagrange : either } \nabla g = \vec{0} \quad (\Rightarrow) \quad \begin{pmatrix} -2x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow (0, 0)$$

but it's
not on curve
 $g(x, y) \neq 0$

$$\text{or} \quad \begin{cases} \nabla T = \lambda \nabla g \\ g = 0 \end{cases}$$

$$\begin{cases} 4x^3 = \lambda(-2x) \\ 4y = \lambda \\ y - x^2 + 3 = 0 \end{cases} \Rightarrow 2x(2x^2 + \lambda) = 0$$

Case I :

$$x = 0$$

$$y = x^2 - 3 = -3$$

$$(0, -3)$$

Case II :

$$2x^2 + \lambda = 0$$

$$\lambda = -2x^2$$

$$y = \frac{\lambda}{4} = -\frac{x^2}{2}$$

$$-\frac{x^2}{2} - x^2 + 3 = 0$$

$$-\frac{3x^2}{2} = -3$$

$$x^2 = 2 \Rightarrow x = \pm\sqrt{2}$$

$$y = -1$$

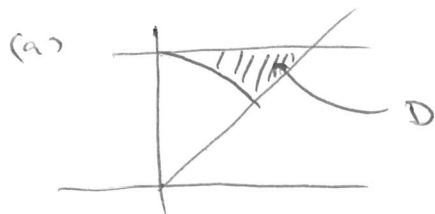
$$(\sqrt{2}, -1) \text{ and } (-\sqrt{2}, -1)$$

$$T(0, -3) = 2 \cdot 9 + 1 = 19$$

$$T(\pm\sqrt{2}, -1) = 4 + 2 + 1 = 7$$

Minimal temp. is 7 at $(\pm\sqrt{2}, -1)$

Problem 7



(b) In polar coordinates

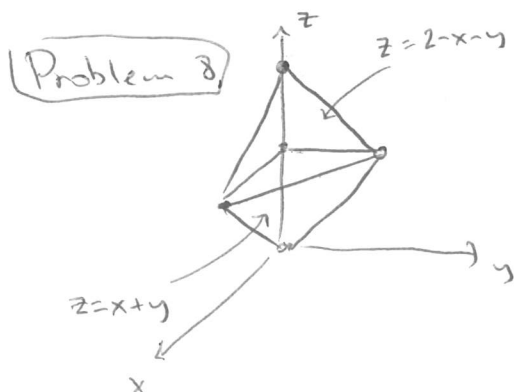
$$\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$$

$$1 \leq r \leq \frac{1}{\sin \theta}$$

↑
since $y \leq 1$ is $r \sin \theta \leq 1$

$$\text{i.e. } r \leq \frac{1}{\sin \theta}$$

$$\begin{aligned} \iint_D \frac{1}{(x^2+y^2)^{3/2}} dx dy &= \int_{\pi/4}^{\pi/2} d\theta \int_1^{\frac{1}{\sin \theta}} \frac{1}{r^3} r dr = \int_{\pi/4}^{\pi/2} d\theta \cdot \left(-\frac{1}{r} \right) \Big|_1^{\frac{1}{\sin \theta}} \\ &= \int_{\pi/4}^{\pi/2} (-\sin \theta + 1) d\theta = (\theta + \cos \theta) \Big|_{\pi/4}^{\pi/2} = \frac{\pi}{2} + 0 - \left(\frac{\pi}{4} + \frac{1}{\sqrt{2}} \right) = \\ &= \frac{\pi}{4} - \frac{1}{\sqrt{2}} \end{aligned}$$



$$\begin{aligned} \iiint_E x dV &= \int_0^1 dx \int_0^{1-x} dy \int_{x+y}^{2-x-y} x dz \\ &= \int_0^1 dx \int_0^{1-x} x (2-x-y-x-y) dy \end{aligned}$$

$$\begin{aligned} &= \int_0^1 dx \int_0^{1-x} (2x - 2x^2 - 2xy) dy = \int_0^1 (2xy - 2x^2y - \frac{2}{2}y^2) \Big|_0^{1-x} dx \\ &= \int_0^1 (2x(1-x) - 2x^2(1-x) - \frac{1}{2}x^2(1-x)^2) dx = \int_0^1 (2x - 2x^2 - 2x^2 + 2x^3 - \frac{1}{2}x^2 + \frac{1}{2}x^3) dx \\ &= \int_0^1 (x - 2x^2 + x^3) dx \\ &= \left(\frac{x^2}{2} - 2\frac{x^3}{3} + \frac{x^4}{4} \right) \Big|_0^1 = \frac{1}{2} - \frac{2}{3} + \frac{1}{4} = \frac{1}{12} \end{aligned}$$